

Linear Grammars

Grammars with
at most one variable at the right side
of a production

Examples:

$S \rightarrow aSb$	$S \rightarrow Ab$
$S \rightarrow \lambda$	$A \rightarrow aAb$
	$A \rightarrow \lambda$

A Non-Linear Grammar

Grammar G :

$$S \rightarrow SS$$
$$S \rightarrow \lambda$$
$$S \rightarrow aSb$$
$$S \rightarrow bSa$$

$$L(G) = \{w : n_a(w) = n_b(w)\}$$



Number of a in string w

Another Linear Grammar

Grammar G :

$$S \rightarrow A$$
$$A \rightarrow aB \mid \lambda$$
$$B \rightarrow Ab$$

$$L(G) = \{a^n b^n : n \geq 0\}$$

Right-Linear Grammars

All productions have form: $A \rightarrow xB$

or

$$A \rightarrow x$$



Example: $S \rightarrow abS$

$$S \rightarrow a$$

string of
terminals

Left-Linear Grammars

All productions have form: $A \rightarrow Bx$

or

$$A \rightarrow x$$

string of
terminals



Example:

$$S \rightarrow Aab$$

$$A \rightarrow Aab \mid B$$

$$B \rightarrow a$$

Regular Grammars

Regular Grammars

A regular grammar is any right-linear or left-linear grammar

Examples:

 G_1

$$S \rightarrow abS$$

$$S \rightarrow a$$

 G_2

$$S \rightarrow Aab$$

$$A \rightarrow Aab \mid B$$

$$B \rightarrow a$$

Observation

Regular grammars generate regular languages

Examples:

G_1

$$S \rightarrow abS$$

$$S \rightarrow a$$

$$L(G_1) = (ab)^* a$$

G_2

$$S \rightarrow Aab$$

$$A \rightarrow Aab \mid B$$

$$B \rightarrow a$$

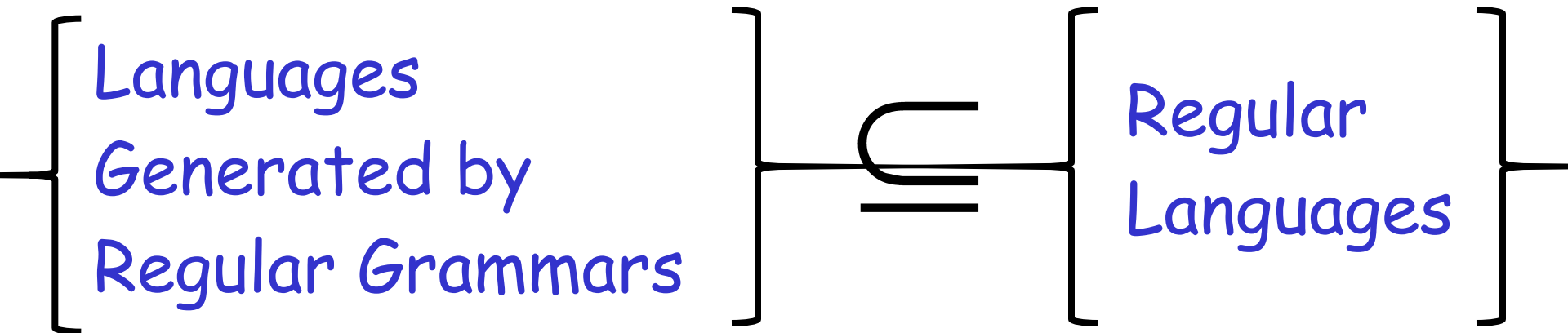
$$L(G_2) = aab(ab)^*$$

Regular Grammars Generate Regular Languages

Theorem

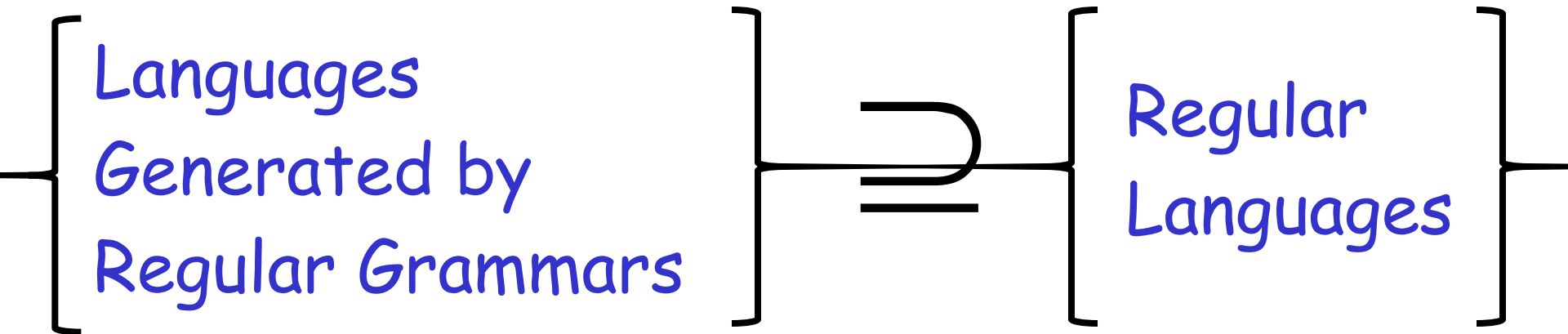
$$\left[\begin{array}{l} \text{Languages} \\ \text{Generated by} \\ \text{Regular Grammars} \end{array} \right] = \left[\begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right]$$

Theorem - Part 1



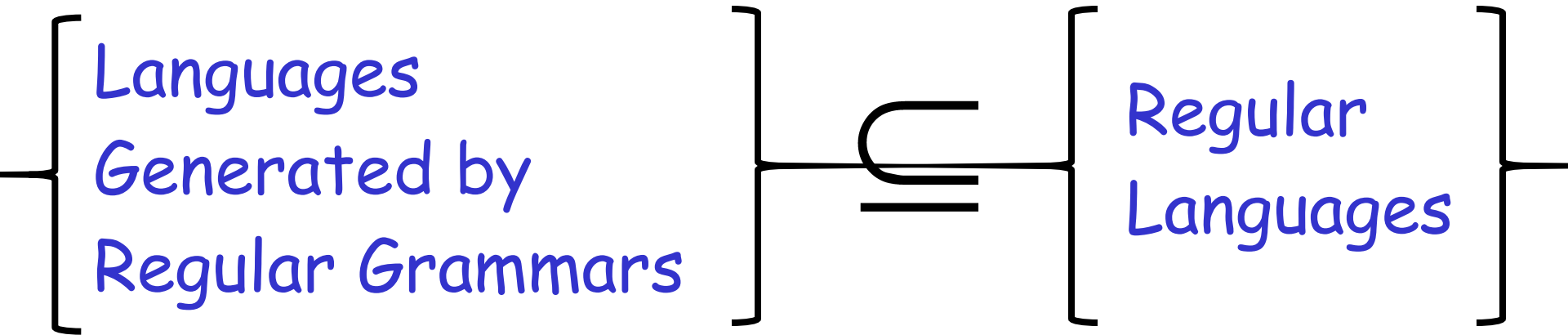
Any regular grammar generates
a regular language

Theorem - Part 2



Any regular language is generated by a regular grammar

Proof - Part 1



The language $L(G)$ generated by any regular grammar G is regular

The case of Right-Linear Grammars

Let G be a right-linear grammar

We will prove: $L(G)$ is regular

Proof idea: We will construct NFA M
with $L(M) = L(G)$

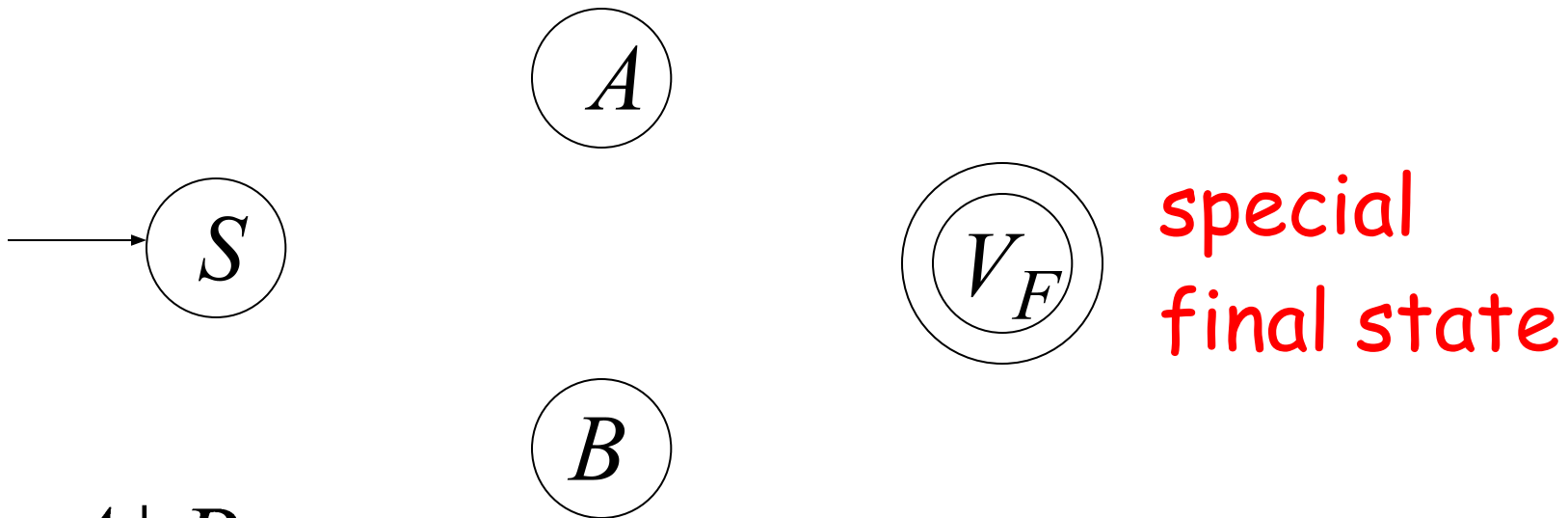
Grammar G is right-linear

Example: $S \rightarrow aA \mid B$

$A \rightarrow aa B$

$B \rightarrow b B \mid a$

Construct NFA M such that every state is a grammar variable:

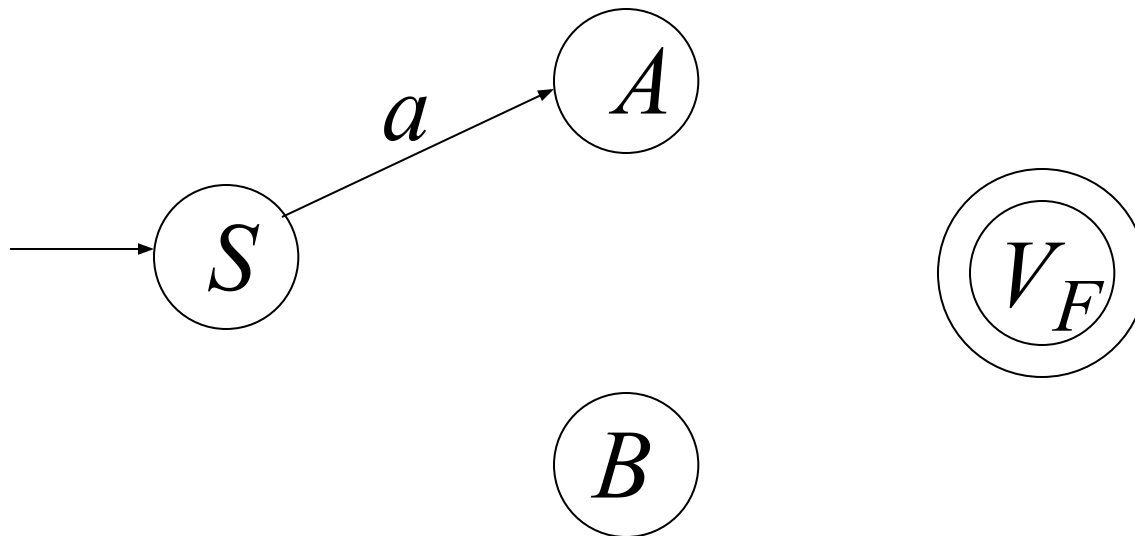


$$S \rightarrow aA \mid B$$

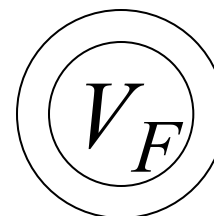
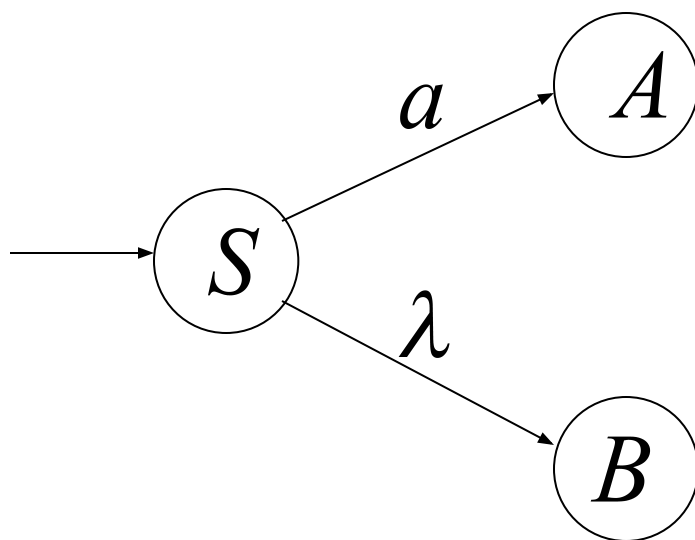
$$A \rightarrow aa B$$

$$B \rightarrow b B \mid a$$

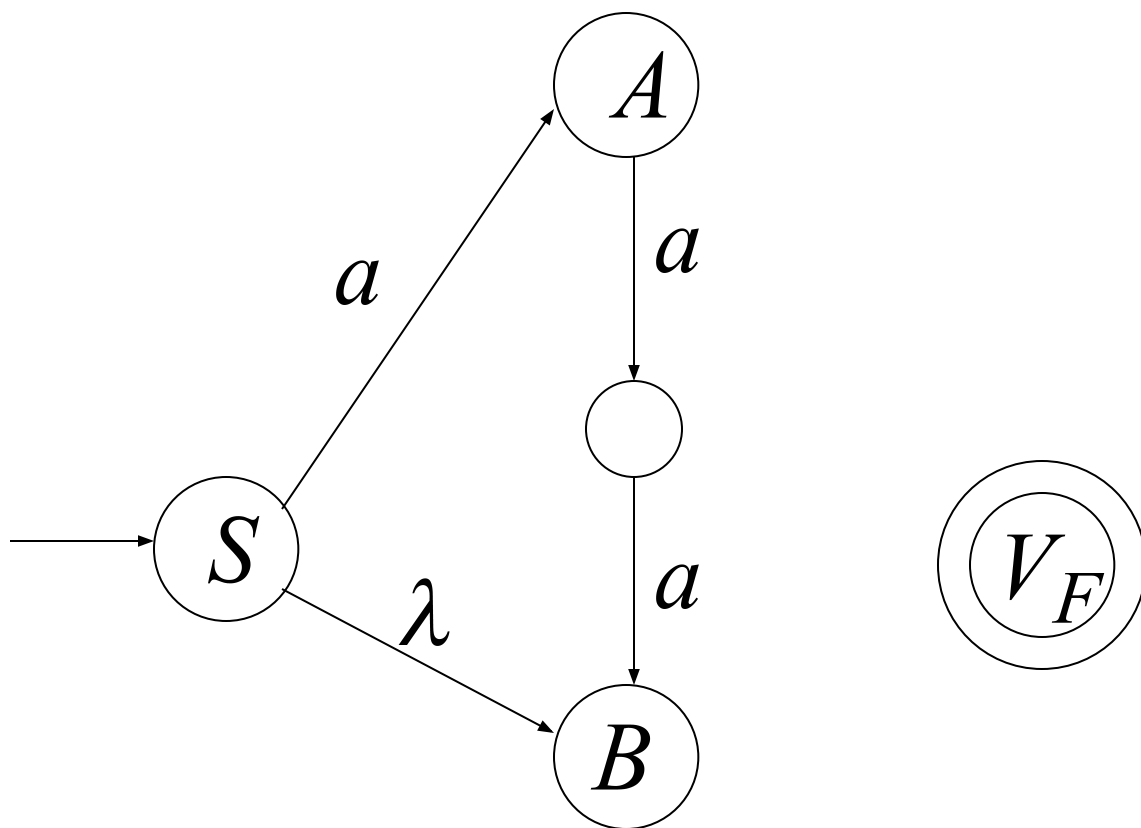
Add edges for each production:



$$S \rightarrow aA$$

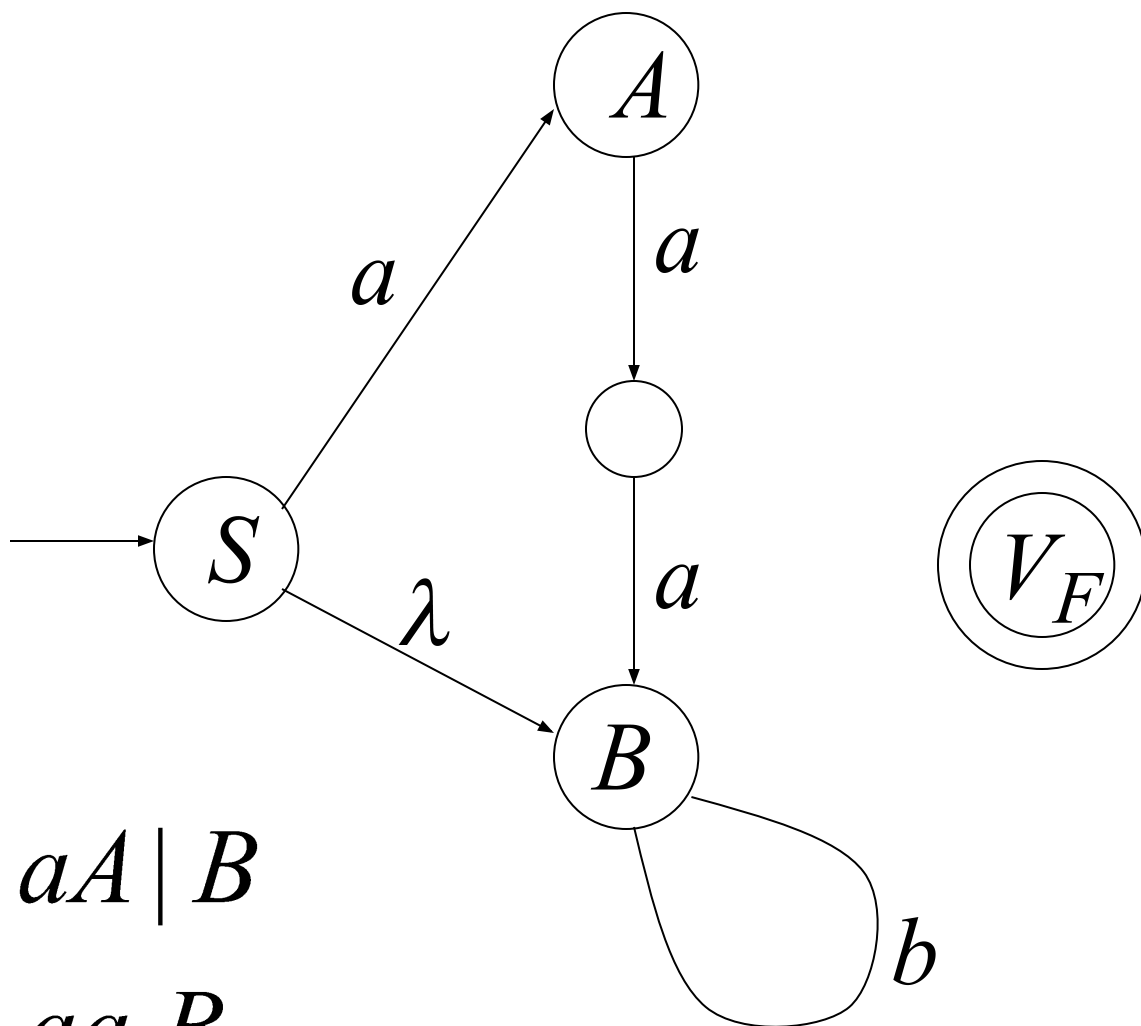


$$S \rightarrow aA \mid B$$



$$S \rightarrow aA \mid B$$

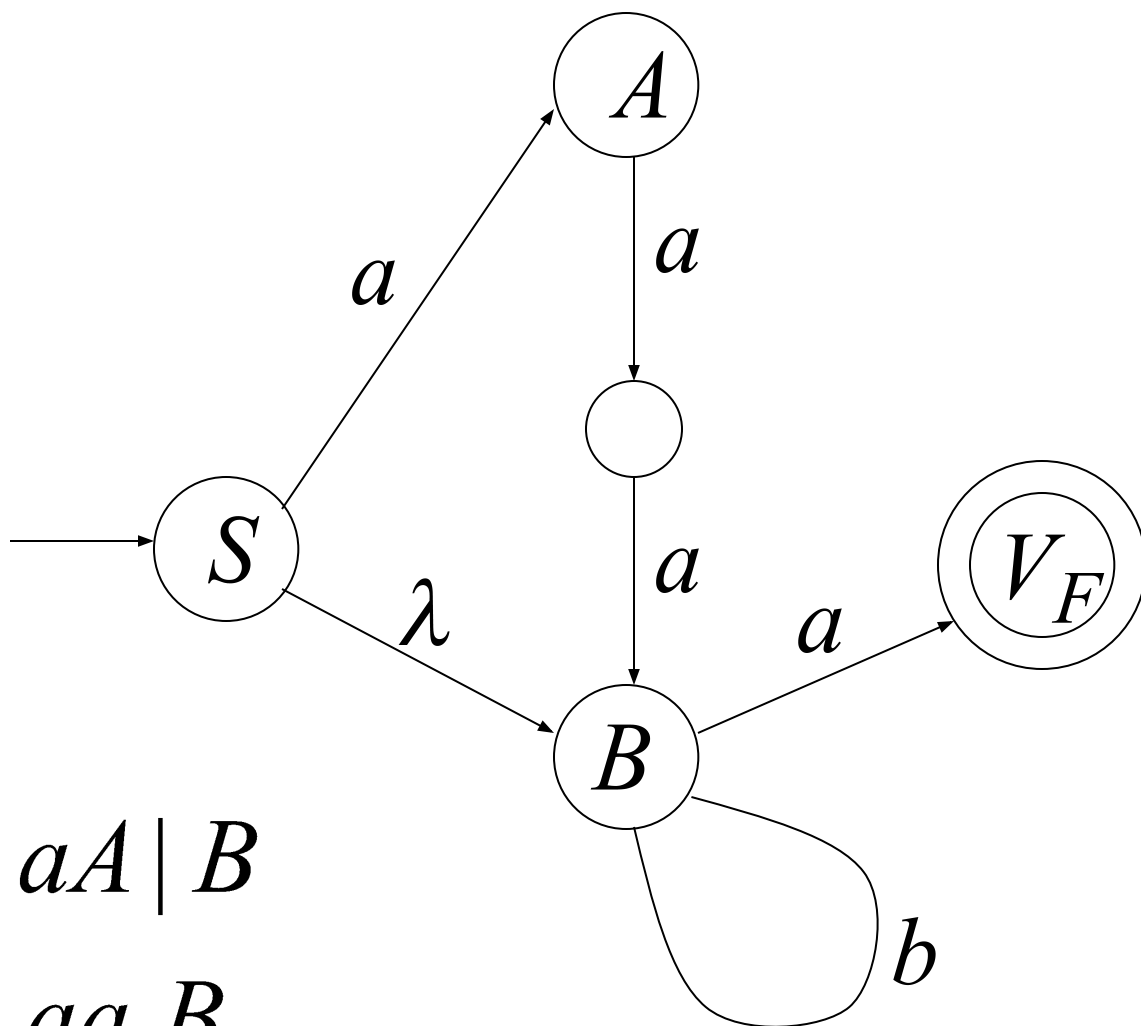
$$A \rightarrow aa B$$



$$S \rightarrow aA \mid B$$

$$A \rightarrow aa B$$

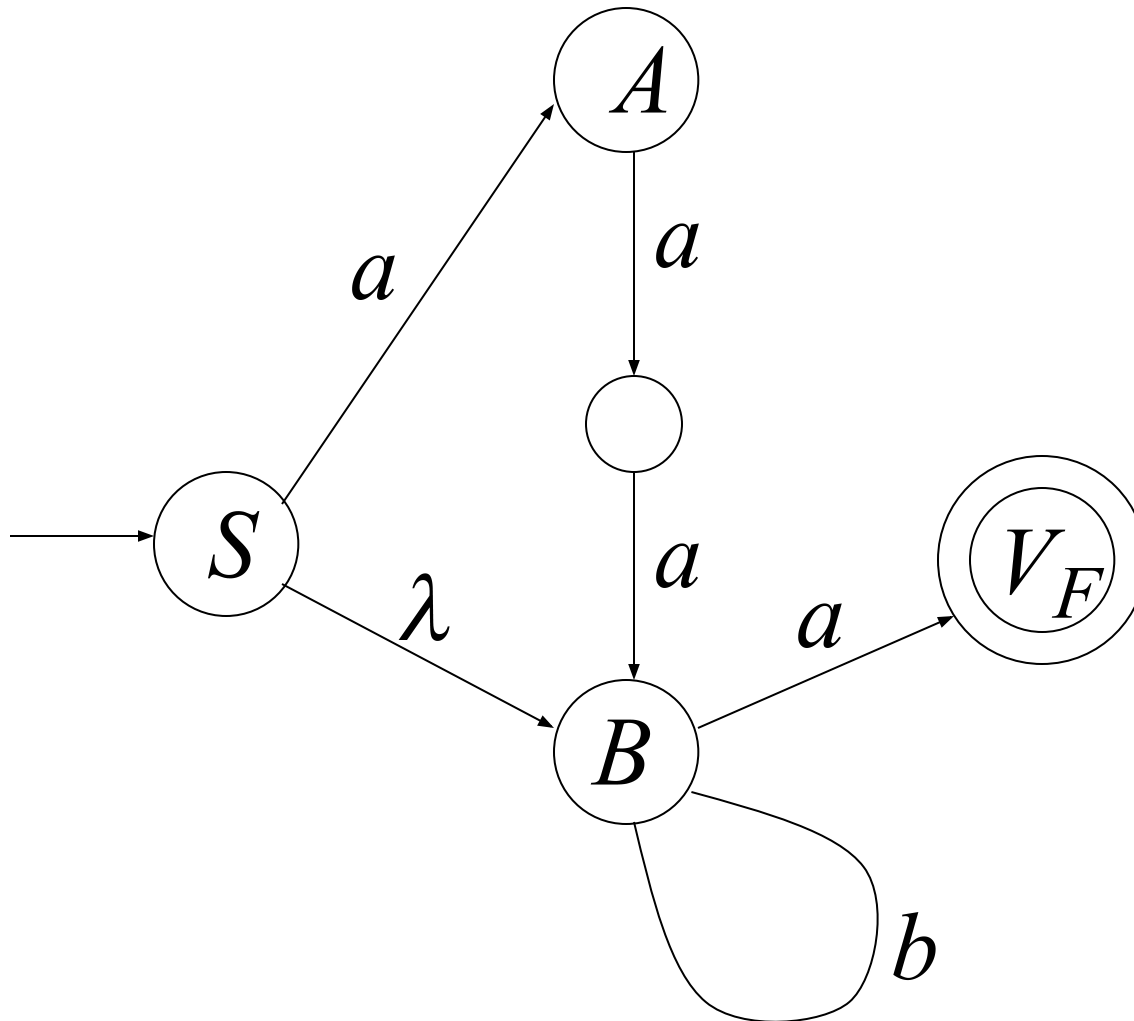
$$B \rightarrow bB$$



$$S \rightarrow aA \mid B$$

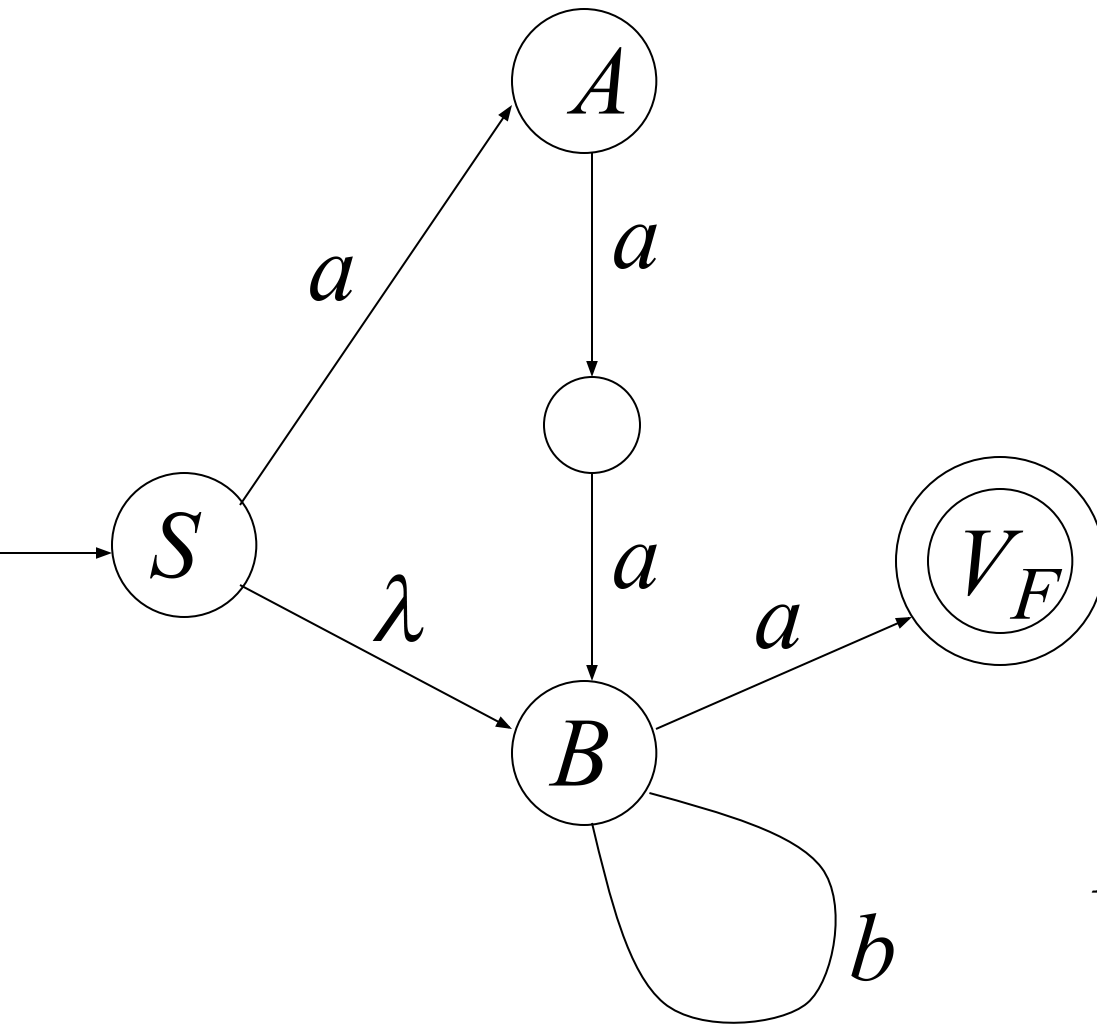
$$A \rightarrow aa B$$

$$B \rightarrow bB \mid a$$



$$S \Rightarrow aA \Rightarrow aaaB \Rightarrow aaabB \Rightarrow aaaba$$

NFA M



Grammar

G

$$S \rightarrow aA \mid B$$

$$A \rightarrow aaB$$

$$B \rightarrow bB \mid a$$

$$L(M) = L(G) = aaab^*a + b^*a$$

In General

A right-linear grammar G

has variables: $V_0, V_1, V_2, \dots$

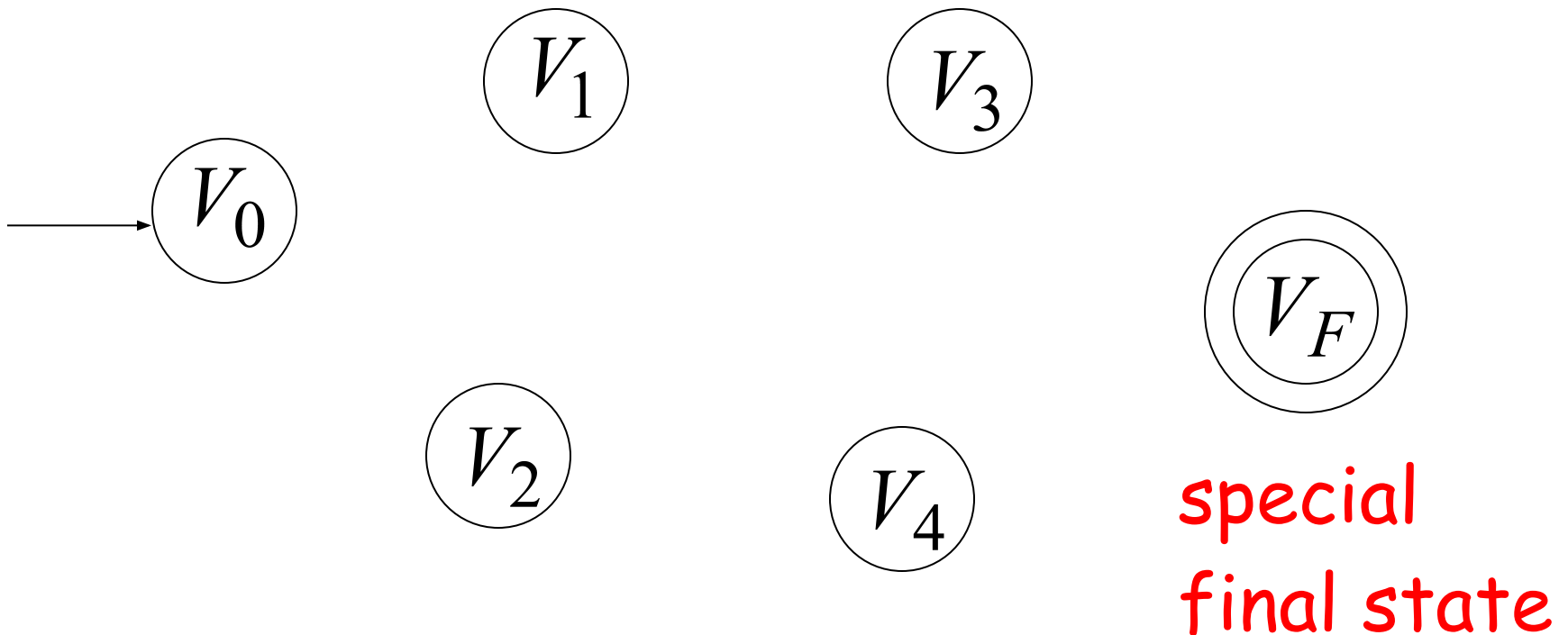
and productions: $V_i \rightarrow a_1 a_2 \dots a_m V_j$

or

$$V_i \rightarrow a_1 a_2 \dots a_m$$

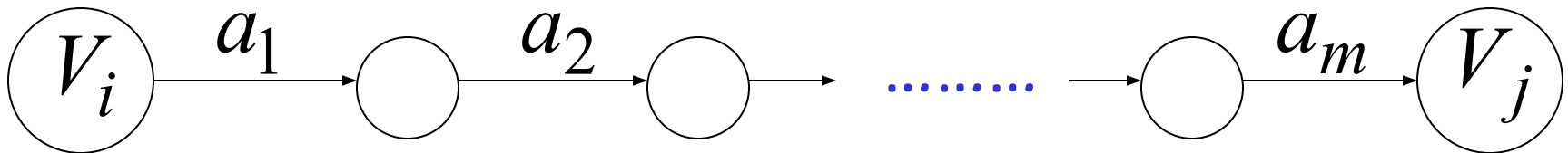
We construct the NFA M such that:

each variable V_i corresponds to a node:



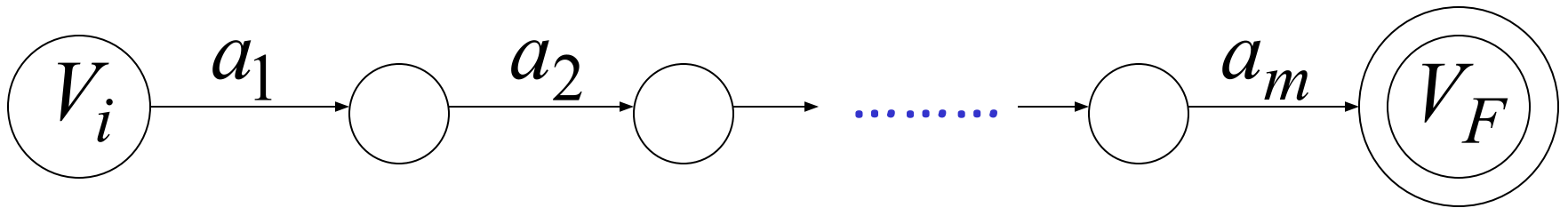
For each production: $V_i \rightarrow a_1 a_2 \boxtimes a_m V_j$

we add transitions and intermediate nodes

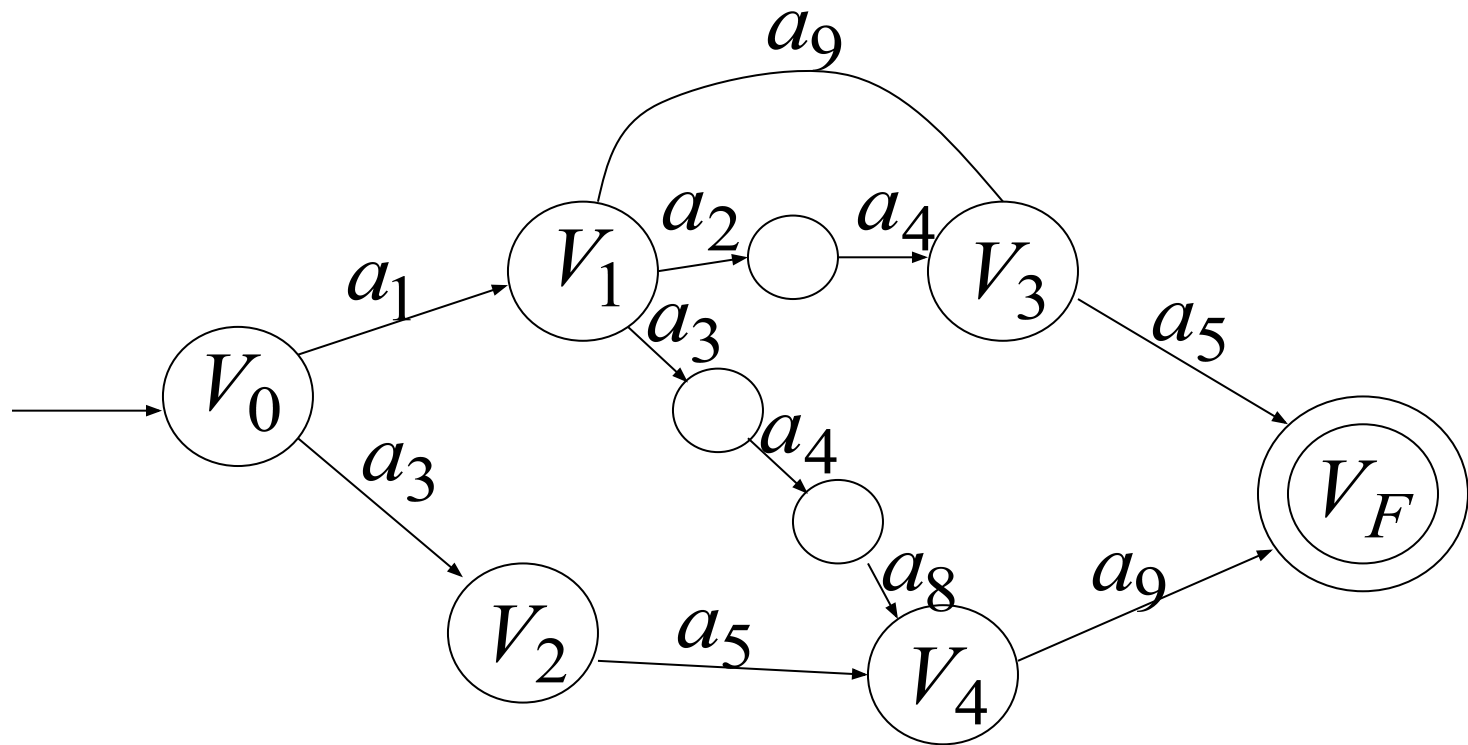


For each production: $V_i \rightarrow a_1 a_2 \boxtimes a_m$

we add transitions and intermediate nodes



Resulting NFA M looks like this:



It holds that: $L(G) = L(M)$

The case of Left-Linear Grammars

Let G be a left-linear grammar

We will prove: $L(G)$ is regular

Proof idea:

We will construct a right-linear grammar G' with $L(G) = L(G')^R$

Since G is left-linear grammar
the productions look like:

$$A \rightarrow Ba_1a_2 \mid a_k$$

$$A \rightarrow a_1a_2 \mid a_k$$

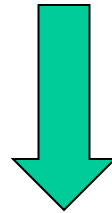
Construct right-linear grammar G'

Left
linear

G

$$A \rightarrow Ba_1a_2 \boxtimes a_k$$

$$A \rightarrow Bv$$



Right
linear

G'

$$A \rightarrow a_k \boxtimes a_2a_1B$$

$$A \rightarrow v^R B$$

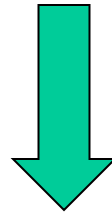
Construct right-linear grammar G'

Left
linear

G

$$A \rightarrow a_1 a_2 \boxtimes a_k$$

$$A \rightarrow v$$



Right
linear

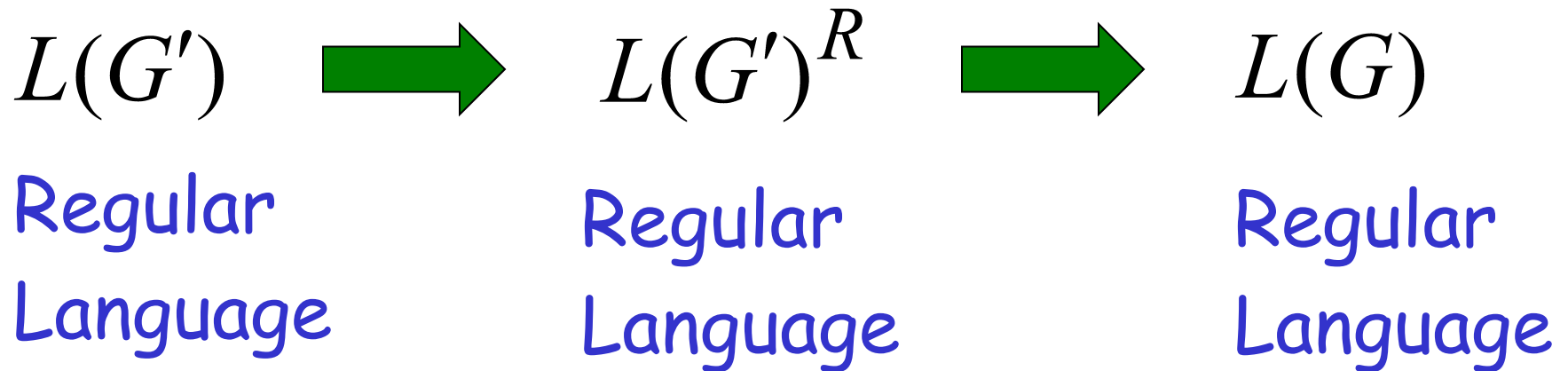
G'

$$A \rightarrow a_k \boxtimes a_2 a_1$$

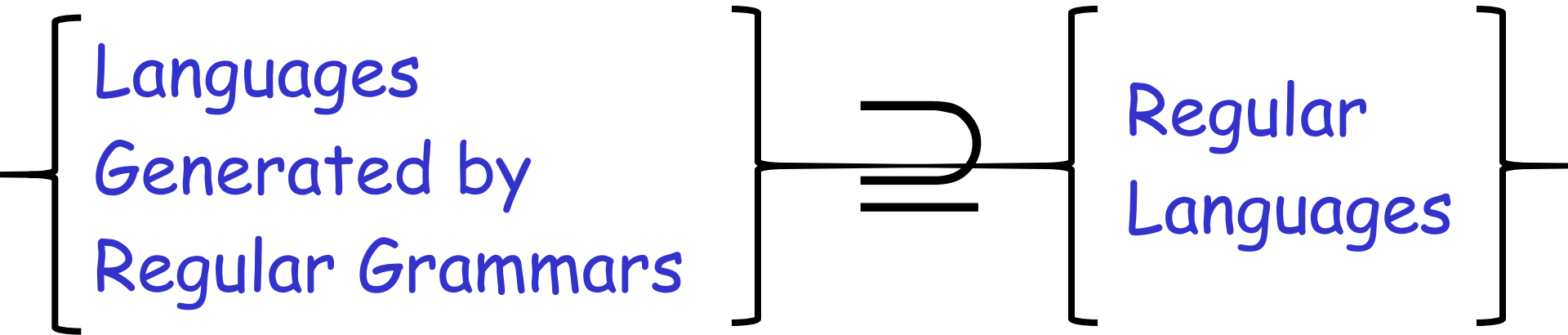
$$A \rightarrow v^R$$

It is easy to see that: $L(G) = L(G')^R$

Since G' is right-linear, we have:



Proof - Part 2



Any regular language L is generated
by some regular grammar G

Any regular language L is generated
by some regular grammar G

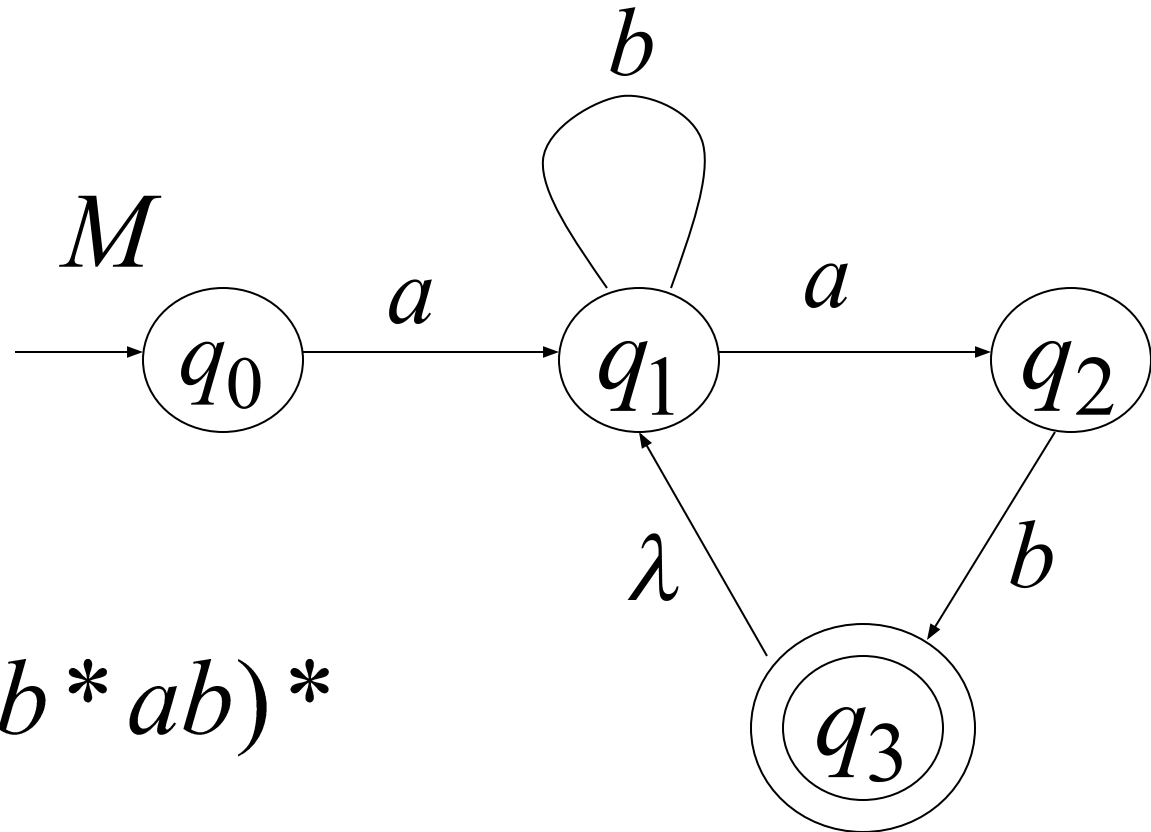
Proof idea:

Let M be the NFA with $L = L(M)$.

Construct from M a regular grammar G
such that $L(M) = L(G)$

Since L is regular
there is an NFA M such that $L = L(M)$

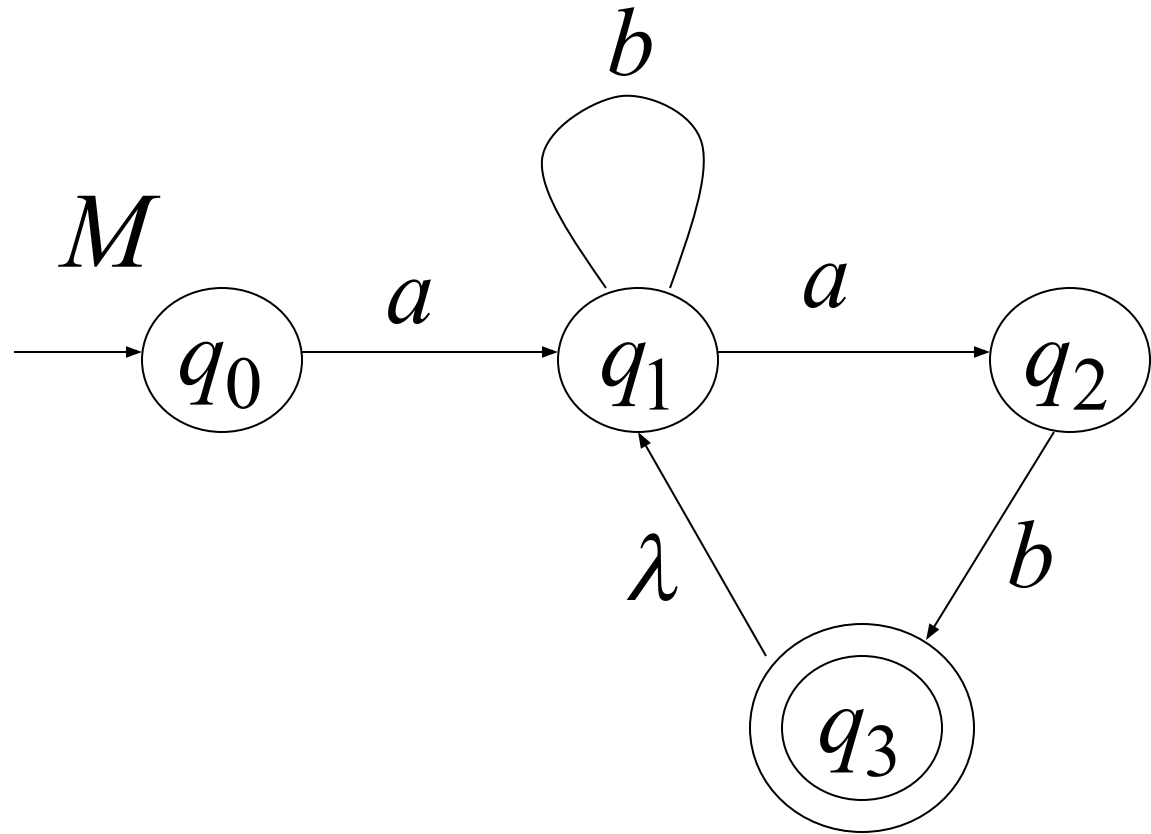
Example:



$$L = ab^*ab(b^*ab)^*$$

$$L = L(M)$$

Convert M to a right-linear grammar

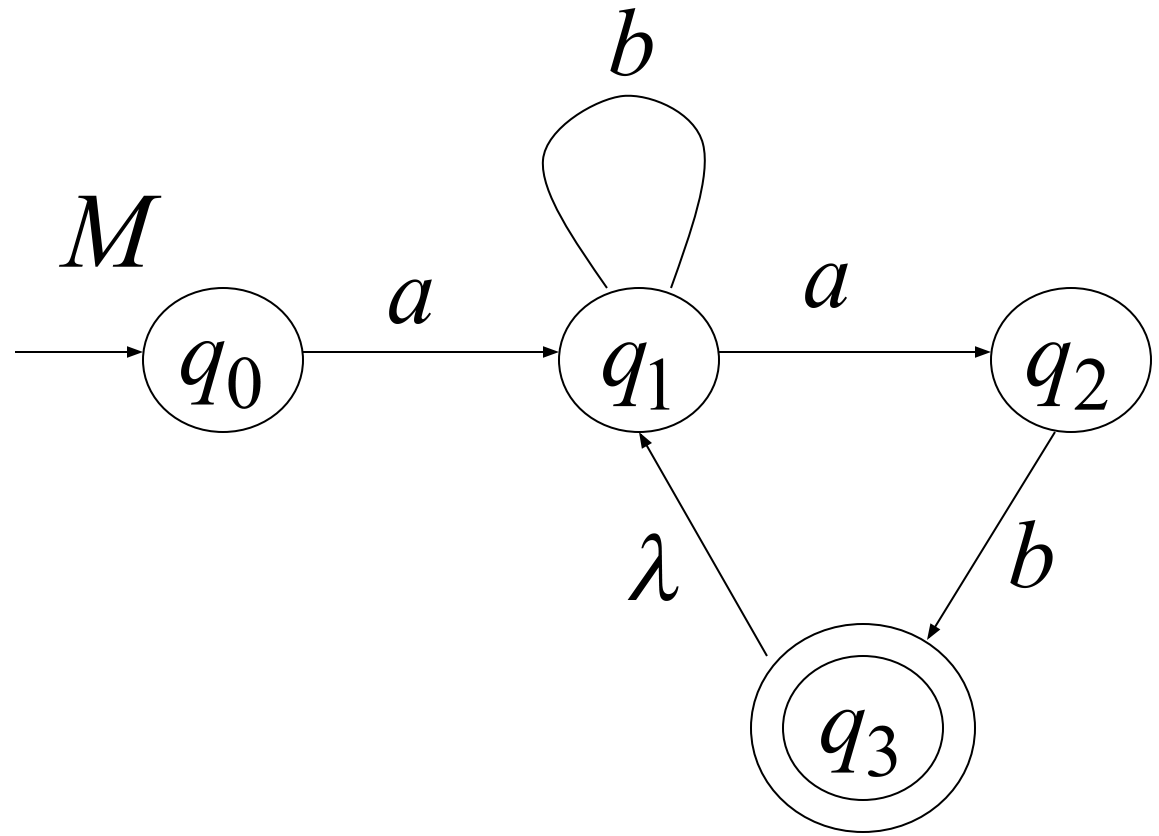


$$q_0 \rightarrow aq_1$$

$q_0 \rightarrow aq_1$

$q_1 \rightarrow bq_1$

$q_1 \rightarrow aq_2$

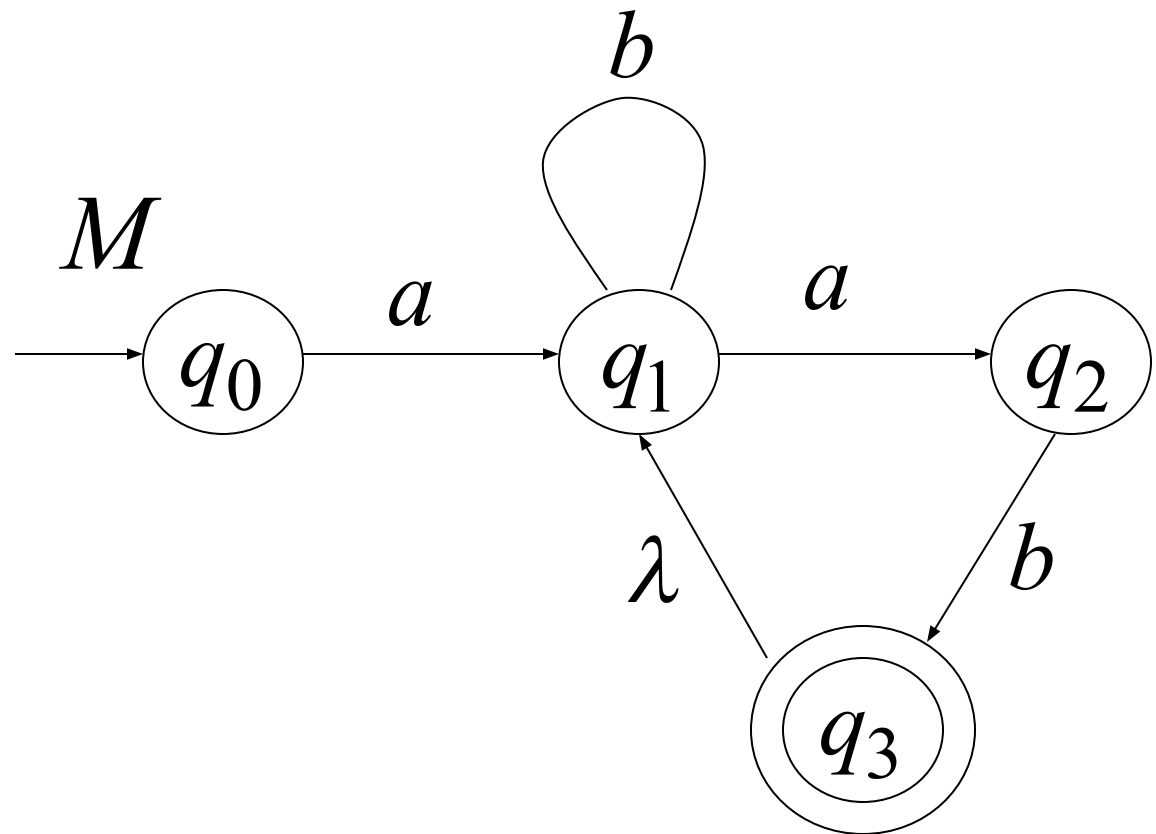


$q_0 \rightarrow aq_1$

$q_1 \rightarrow bq_1$

$q_1 \rightarrow aq_2$

$q_2 \rightarrow bq_3$



$$L(G) = L(M) = L$$

G

$$q_0 \rightarrow aq_1$$

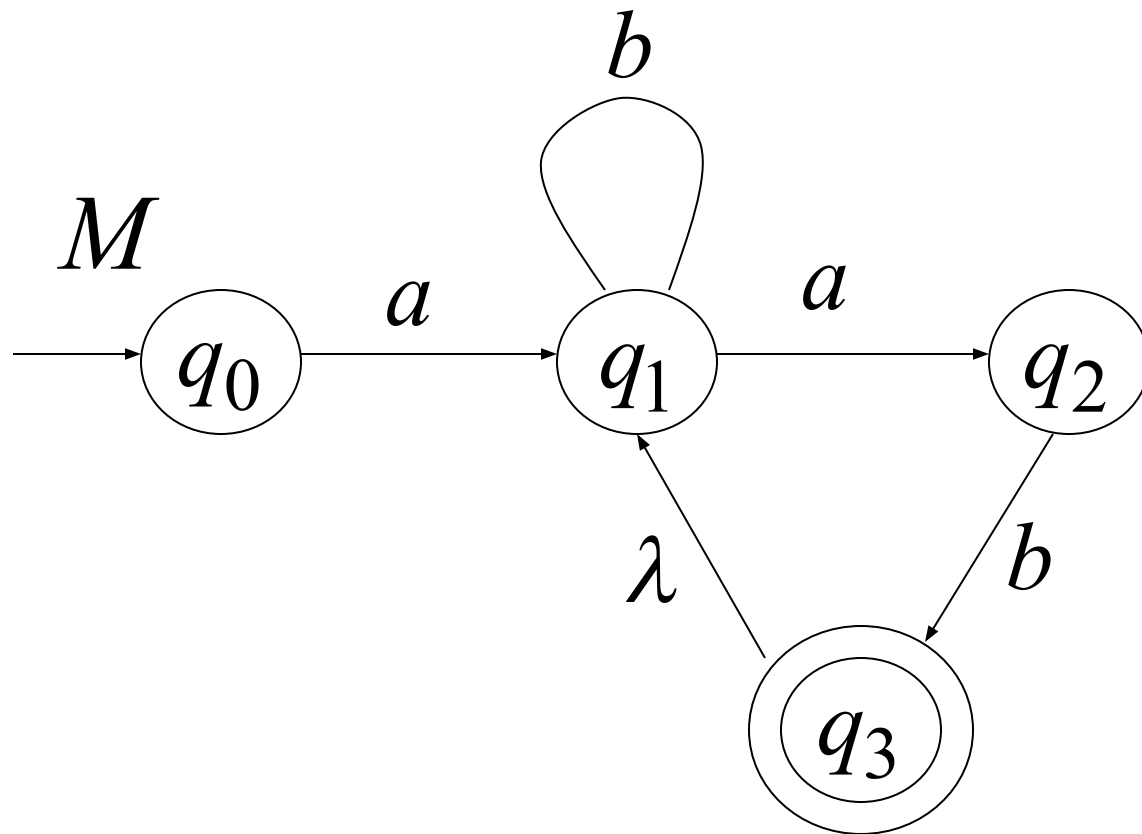
$$q_1 \rightarrow bq_1$$

$$q_1 \rightarrow aq_2$$

$$q_2 \rightarrow bq_3$$

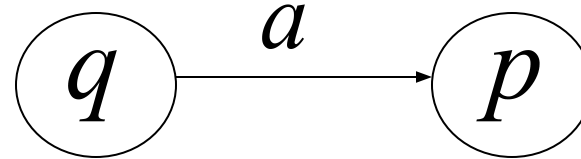
$$q_3 \rightarrow q_1$$

$$q_3 \rightarrow \lambda$$

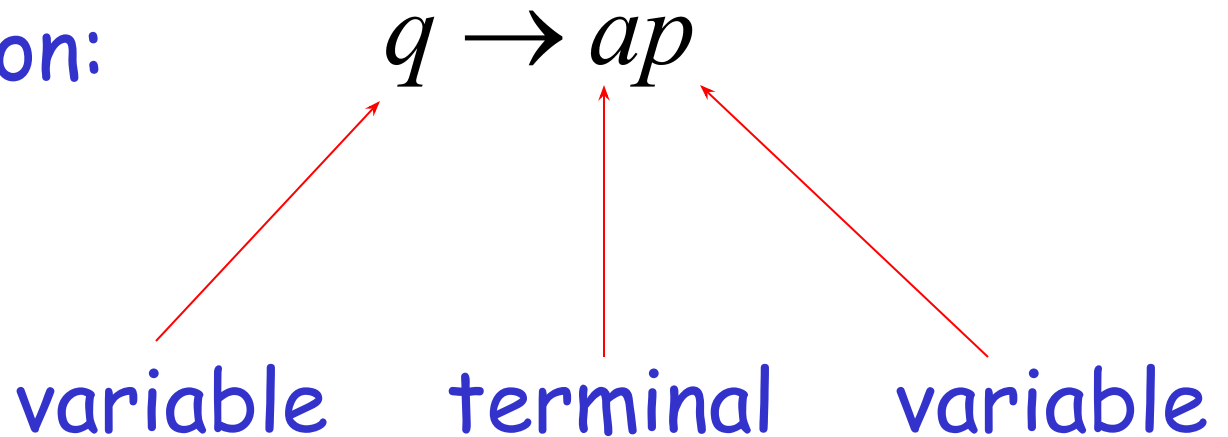


In General

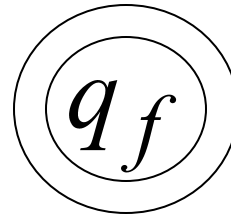
For any transition:



Add production:



For any final state:



Add production:

$$q_f \rightarrow \lambda$$

Since G is right-linear grammar

G is also a regular grammar

with $L(G) = L(M) = L$

Pumping lemma

- This theorem can be used to prove that certain sets are not regular.
- We now give the steps needed for proving that a given set is not regular.

Step 1: Assume L is regular. Let ' n ' be the number of states in the corresponding F.A.

Step 2: choose a string w such that $|w| \geq n$. Use pumping lemma to write $w = xyz$, with $|xy| \leq n$ and $|y| > 0$

Step 3: Find suitable integer ' i ' such that $xy^i z$ should not belong to L .

Pumping Lemma

• Pumping Lemma is used to prove that a Language is NOT REGULAR

• It cannot be used to prove that a Language is Regular

If A is a Regular Language, then A has a Pumping Length ' P ' such that any string ' S ' where $|S| \geq P$ may be divided into 3 parts $S = x y z$ such that the following conditions must be true:

- (1) $x y^i z \in A$ for every $i \geq 0$
- (2) $|y| > 0$
- (3) $|xy| \leq P$

To prove that a language is not Regular using PUMPING LEMMA, follow the below steps:

(We prove using Contradiction)

- > Assume that A is Regular
- > It has to have a Pumping Length (say P)
- > All strings longer than P can be pumped $|S| \geq P$
- > Now find a string ' S ' in A such that $|S| \geq P$
- > Divide S into $x y z$
- > Show that $x y^i z \notin A$ for some i
- > Then consider all ways that S can be divided into $x y z$
- > Show that none of these can satisfy all the 3 pumping conditions at the same time
- > S cannot be Pumped == CONTRADICTION

Example 1

Show that the set, $L = \{a^n \mid n \geq 0\}$ is regular or not.

$W = aa$

Case 1:

$X = \text{null}$

$Y = aa$

$Z = \text{null}$

For all value of i , still belongs to language.

Example 2

Show that the set, $L = \{a^{2n} \mid n \geq 0\}$ is regular or not.

$W = aa$

Case 1:

$X = \text{null}$

$Y = aa$

$Z = \text{null}$

For all value of i , still belongs to language.

Example 3

Show that the set, $L = \{a^n b^n \mid n > 0\}$ is regular or not.

$W = ab$

Case 1:

$X = a$

$Y = b$

$Z = \text{null}$

For all value of i ,

When $i=1$ result will be still belongs to language.

When $i=2$ result will be abb , string not belongs to language, hence proved language is not regular

Show that the set, $L = \{a^n b^n c^n \mid n > 0\}$ is regular or not.

$W = abc$

Case 1:

$X = a$

$Y = b$

$Z = c$

For all value of i ,

When $i=1$ result will be still belongs to language.

When $i=2$ result will be $abbc$, string not belongs to language, hence proved language is not

Pumping lemma

Apply Pumping Lemma on languages **L1**, **L2** and **L3** to prove that these are not regular languages.

$$(i) \quad L1 = \{0^n 1^m : n \leq m+3\}$$
$$110(10)^n : n \geq 0\}$$

$$(ii) \quad L2 = \{01(1100)^n$$

$$(iii) \quad L3 = \{a^i b^j c^k : i \geq j \geq k \geq 1\}$$

Pumping lemma

Is the language

$L = \{w \mid w \text{ is a binary string with at least as many 1's as 0's}\}$

regular? Prove your answer using pumping lemma

Bilal, a student of CS301, has submitted following answer:

I will prove that L is not regular, using the pumping lemma.

i. Let p is a pumping length

ii. Let $w = 01^p$, so that $w \in L$ and $|w| \geq p$.

iii. Let $w = xyz$, with $x = \epsilon$, $y = 0$, and $z = 1^p$, so that $|xy| \leq p$ and $|y| > 0$.

iv. Then $xy^{p+1}z$ is not in L .

Identify the single incorrect line in Bilal's proof and explain what he did wrong.

(NOTE: do not list small mistakes, typos, etc.; explain the major logical error which invalidates Bilal's proof.)

Solution

In line iii., Bilal did wrong split of w . As 0 is finite in w , therefore cannot be pumped and hence invalidate

the proof. In another words, Bilal is not permitted to pick a specific split of w .

b. "Let L be any finite, nonempty language of binary strings. Is L regular?"

Jameel has submitted the following answer:

I will prove that L is not regular, using the pumping lemma.

i. Since L is finite, it has a longest string. Let p be the length of the longest string in L .

ii. Let $w \in L$, such that $|w| \geq p$.

iii. Let $w = xyz$, so that $|xy| \leq p$ and $|y| > 0$.

iv. Then xy^2z is not in L , since it has length longer than p , but all the strings in L have length at most p .

Identify the single incorrect line in Jameel's proof, and explain what he did wrong.

(NOTE: Do not just say, " L is actually regular"; explain the major logical error which invalidates Jameel's proof.)

